



Research article

Trajectory tracking control of a coaxial rotor drone: Time-delay estimation-based optimal model-free fuzzy logic approach

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ABSTRACT

This paper proposes a control algorithm for controlling the position and attitude of a coaxial rotor drone without knowing the model dynamics. To overcome the major drawback of model-dependent approaches, an optimal model-free fuzzy controller (OMFFC) based on the estimation of the unknown dynamic function of the system is proposed. A time-delay estimation (TDE) technique is effectively exploited to approximate the unknown dynamic function of the system. The estimation error is then offset using a robust adaptive fuzzy logic compensator. Based on Lyapunov stability arguments, the global asymptotic stability of the coaxial rotor drone system is proven. Moreover, a flower pollination-based algorithm is also proposed to generate the optimal parameters to address the trade-off between optimal tracking performance and the design conditions related to the closed-loop stability requirements. The numerical simulations illustrate how the proposed methodology leads to the best performance, as well as less computational complexity compared to the standard proportional–integral–derivative and time-delay estimation-based controllers in the presence of external disturbances.

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1. Introduction

Unmanned aerial vehicles (UAVs) have attracted considerable attention due to their use in numerous civil and military applications such as entertainment and sport, search and rescue operations, agriculture, police and traffic surveillance, and more recently chemical spraying [1,2]. Among the different types of UAVs, coaxial rotor helicopters have many attractive advantages such as a small compact structure, vertical take-off and landing (VTOL) capabilities, and excellent hovering performance. Moreover, coaxial rotors exhibit better thrust and have the ability to operate in multiple domains making them extremely useful [3]. Their dynamics are characterized by a strong coupling mechanical structure, high nonlinearities, multidimensional underactuated system, and unstable in an open loop with non-minimum phase probability. These flying machines that can be fully autonomous or controlled remotely are subjected to various physical constraints, parameter uncertainties, and external disturbances such as internal friction, mass imbalance, payload, and wind gusts.

For these reasons, the design of robust flight tracking control is difficult.

1.1. Related works on coaxial rotor

Like with quadrotors [4,5], two main purposes are considered for controlling coaxial rotors: attitude control related to existing rotational angles, and position control assigned to maintain the coaxial in a predefined position. Many linear and nonlinear control strategies have been proposed in the literature for controlling coaxial rotors such as proportional–integral–derivative controllers (PID) [6], linear quadratic (LQ) regulators [7], H_∞ control [8], feedback linearization control [9], backstepping control (BC) [10] and sliding-mode control (SMC) [11]. Linear control approaches are based on a linear model of coaxial rotors and only operate effectively near the operating points. The nonlinear control approaches have a fast convergence rate and they can handle external disturbances more efficiently. However, they have some hardness such as chattering and high sensitivity to measurement noise in pure SMC, singularity problem and complexity issues with the backstepping control technique.

Therefore, to overcome one or more of these drawbacks, hybrid control strategies combining different approaches to improve

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the control performance have been reported in many publications. For example, in [12], a linear rate-bounded PID controller was combined with SMC approach for a fully-actuated coaxial rotor subsystem while the under-actuated part was controlled with a SMC controller alone. In [13], model predictive control (MPC) and SMC were applied for motion control of a coaxial aerial underwater vehicle to enable cross-domain locomotion of the vehicle in the presence of environmental disturbances. An adaptive compensation control scheme based on a backstepping dual-loop structure was proposed in [14] for coaxial rotor vehicles to perform trajectory tracking. Adaptive fault-tolerant control based on a backstepping approach was investigated in [15] to offset the effect caused by actuator failures. A common characteristic of these controllers is that the design requires accurate knowledge of the system dynamics, which is rather difficult in practical applications. In addition, the stability of coaxial rotor drone systems in the path-following is not always proven, leading to decrease the reliability of the proposed controllers. As a solution, fuzzy logic control (FLC) [16] and neural network control have been investigated with the previous approaches [17]. However, the characterization of such intelligent networks with continuous online learning is rather difficult and increases the complexity of the system. Furthermore, the design parameters are often selected randomly and/or require expert knowledge, thus reducing their accuracy and making them difficult to implement.

Oriented to solving the model dependency problem, a model-free control (MFC) approach has been recently employed to deal with the requirement of an accurate dynamic model and to make the control law dependent only by real-time measurements [18]. Surprisingly, very few studies related to this topic can be found in the open literature, as in [19–30]. Based on algebraic methods, the authors in [19,20] proposed the first MFC control schemes for driving quadrotor UAVs. These control strategies focus their approach on local models which are defined in a short time where the stability of the proposed control laws is only proven locally. Furthermore, no specific criteria are defined to select the design parameters, which is a considerable limitation of these approaches. In [21,22], the authors proposed a MFC approach based on the SMC method with adaptive proportional–derivative action. The major drawback of this method is the chattering effect. Robust MFCs based on a backstepping technique for quadrotor and 12-rotor UAVs were investigated in [23,24], where the unknown dynamic functions were approximated using adaptive nonlinear estimators. These approaches are based on many derivatives of the reference and measured variables making the control system extremely complex, specifically with the presence of the measurement noise. This drawback is resolved in [25] via an approximation-free controller considering the thrust saturation case. In [26–28], adaptive model-free fuzzy controllers for multi-rotor UAVs were developed taking into account the failures on actuators and sensors. The results of these studies show that the control methods can effectively deal with faults and unknown dynamics with good tracking performance. However, they introduce complex updating algorithms to solve only fault control issues without considering external disturbances. An adaptive fuzzy PD-based MFC strategy for 3D trajectory tracking of the quadrotor without proof of stability is addressed in [29]. The tracking problem of a quadcopter was investigated in [30], using a type-2 fuzzy neural network (T2FNN)-based adaptive model-free controller. The simulation results show that this control method achieves good performance but its implementation in real systems is difficult as it introduces a heavy hybrid learning algorithm.

Compared with other MFC strategies, the TDE-based control technique has several merits that make it more suitable for many practical applications [31–33]. This approach, used to approximate the nonlinear function of the system, only requires the

measured input/output data of the controlled system to calculate the law. Thus, no partial or total prior knowledge of the system dynamics or training processes is needed for TDE-based MFC design [34]. Under some practical assumptions, the asymptotic convergence and stability of the TDE-based control system can be guaranteed. Many studies proposed in the literature show that the TDE-based control approach is easy to implement in real applications with a small computational burden and it is highly robust with respect to system uncertainties and disturbances [35]. In this work, the TDE technique was employed to estimate the coaxial dynamics and system disturbances that are unknown, and then to develop a robust model-free control scheme.

1.2. Proposed methodology and contributions

In this paper, we propose an optimal model-free fuzzy controller (OMFFC) to control the attitude and position of a coaxial rotor drone. The proposed approach contains three elements: PID and TDE controllers coupled with a fuzzy logic compensator. The PID controller, a simple control unit, adapted to real applications and industrial processes, was employed to reach the best performance through the optimal off-line setting of its design parameters. The TDE technique was applied to estimate all the unknown system dynamics and disturbances of the coaxial rotor, which improves the robustness of the proposed controller and maintains its simplicity. The fuzzy control part was added to deal with TDE-errors and to guarantee the system stability in the Lyapunov sense. The fuzzy compensator was then introduced with a simple configuration to further reduce system complexity and simplify its implementation. Finally, we introduce an optimization method based on the flower pollination algorithm (FPA) to overcome the problem regarding the selection of the design parameters and achieve optimal behavior of the proposed control approach [36,37]. Compared with other studies, the particular features of the proposed optimal model-free control solution can be highlighted as follows:

- The model-based control approaches proposed for multi-rotor UAVs, as in [6,10,12], assume that the aerial vehicle model is partially or totally known for the control design, therefore resulting in high sensitivity to the unmodeled dynamics and the effect of parameter variations, and considerably limiting their range of implementation in practical applications. Instead, the proposed approach is model-free and does not need any prior information of the system parameters and its dynamic functions. This allows good performance to be achieved and nonlinearity and uncertainty control problems to be overcome.
- Intelligent MFC control methods based on fuzzy logic systems and/or neural networks have been the most investigated in the literature regarding the position and attitude control of multi-rotor aircraft [26–28,30]. The design of the controllers in these studies is complex and requires regularly updated expert knowledge with heavy adaptive algorithms requiring a lot of computations. They are, therefore, difficult to implement in real-time, especially when the control laws are restricted by assumptions that are tricky to verify on real physical systems. In the present paper, a simple TDE technique is employed instead to estimate all the unknown dynamics and disturbances of coaxial rotor, while the fuzzy logic component is introduced here to deal with approximation errors using a simple design architecture. The proposed MFC scheme has a low-complexity design with few computations using input/output measurements making it easy to implement in real-time without any constraints regarding the design parameters.

- The stabilization is addressed in various MFC strategies developed to drive quadrotor drones [19–21] and the results show that path tracking is achieved with good performance. However, they employ ultra-local models to design robust controllers; this remains insufficient as only the local stability of the system is validated. Conversely, in the proposed control approach, the stability and the asymptotic convergence of the tracking errors to zero of the individual subsystems and the complete coaxial rotor system are demonstrated through a rigorous proof of stability using the Lyapunov theory.
- Throughout the design analysis of most control approaches developed for UAVs, as in [13,19,26,28], several design parameters emerge which have to be chosen properly. These design parameters are generally selected using the trial-and-error procedure, which should meet the stabilization requirements of the control system. This exhaustive method does not lead to an optimal solution in terms of best performance which conduct to greater limitations of the proposed approaches. Consequently, in the present work, the selection of the design parameters of the proposed controller is converted into an optimization problem to ensure the best performances. As a modern optimization technique, meta-heuristic FPA is employed to provide a systematic procedure to find the optimal values of the control parameters.

To summarize, this paper investigates the design of an OMFFC for a coaxial rotor vehicle, which claims the following contributions:

- A novel model-free control approach is suggested for position and attitude tracking control of a coaxial rotor UAV thus resulting in low sensitivity to the unmodeled dynamics, parameter variations, and disturbances.
- A hybrid control strategy based on the TDE approach and an adaptive fuzzy logic compensator is proposed to improve the robustness of the control system with low computational complexity and to ensure easy implementation in real applications.
- Based on Lyapunov stability arguments, we demonstrate the asymptotic stability of the whole system, which provides the adaptive laws to guarantee the asymptotic convergence of the tracking errors.
- A flower pollination optimization algorithm is introduced to generate the optimal PID parameters to deal with the trade-off between the optimal tracking performance and the design conditions related to the closed-loop stability requirements.

The rest of this manuscript is outlined as follows. Section 2 describes the mathematical model of the coaxial rotor drone and the corresponding problem statement used for control purposes is formulated. In Section 3, the design of the OMFFC based on the TDE technique, an adaptive fuzzy system, and an FPA algorithm are proposed with closed-loop stability analysis. Section 4.1 compares the performance of the OMFFC designed with other related controllers such as the standard PID, the MFFC, and the TDEC through numerical simulation. Finally, the conclusion of this study and future research are presented in Section 5.

2. Model description and problem statement

The coaxial rotor helicopter has six degrees of freedom (6-DOF) and is an underactuated system characterized by a strong coupling mechanical structure driven by two brushless DC motors turning in opposite directions with Swash-plate incidence angles and two servo motors placed on the lower rotor [10]. The first step to model the dynamics is to consider the vehicle as a solid body that can move in a 3D space.

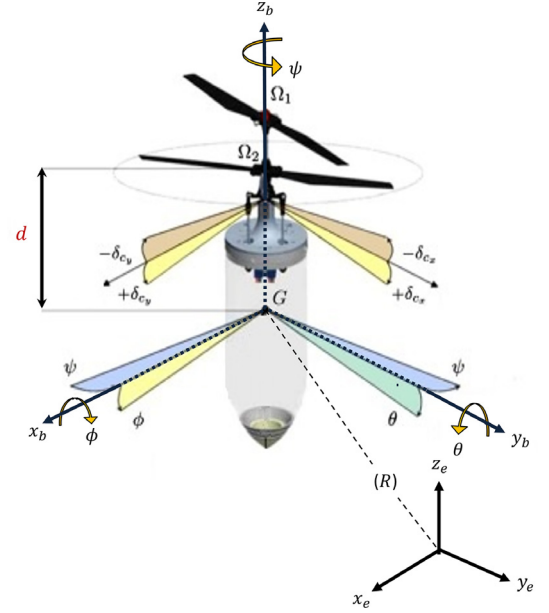


Fig. 1. Coaxial rotor vehicle configuration.

2.1. Dynamics model

To study in 3D space the movement of the coaxial rotor vehicle incorporating a force and moment generation process, let us introduce two frames: the inertial (Earth) reference frame $E(o_e, x_e, y_e, z_e)$ and the body reference frame $B(o_b, x_b, y_b, z_b)$ where the origin of the body reference frame is located at the center of gravity G of the coaxial rotor. Fig. 1 shows the schematic representation of the helicopter. The movement of the drone in the inertial frame is achieved by translating the coordinates $P = [x, y, z]^T$, and the orientation is given by the Euler angles $\eta = [\phi, \theta, \psi]^T$, where ϕ , θ and ψ denote the roll angle around the x -axis, the pitch angle around the y -axis, and the yaw angle around the z -axis, respectively. The position and rotation of the coaxial rotor are controlled via the thrust T and the three control torques $(\tau_\phi, \tau_\theta, \tau_\psi)$. The transition from the frame $E(o_e, x_e, y_e, z_e)$ to the frame $B(o_b, x_b, y_b, z_b)$ is obtained by applying the rotation matrix [10]

$$R = \begin{pmatrix} c\theta c\psi & s\theta c\psi & -c\phi s\psi & c\phi s\theta c\psi + s\phi s\psi \\ c\theta s\psi & s\theta s\psi & c\phi c\psi & c\phi s\theta s\psi - s\phi c\psi \\ -s\theta & c\theta & 0 & c\phi c\theta \end{pmatrix} \quad (1)$$

where s^* and c^* are the shorthand notations of $\sin^*$ and $\cos^*$, respectively. The dynamic model of the coaxial rotor system can be expressed using a Newton–Euler formulation, as in [15]

$$\begin{cases} m\ddot{P} = RT - mg(0, 0, 1)^T + F_{ext} \\ J\ddot{\eta} = \Psi_\eta^T \tau - C(\eta, \dot{\eta})\dot{\eta} + \tau_{ext} \end{cases} \quad (2)$$

where g is the acceleration due to a gravity, m is the body mass of the drone, J is a symmetric and positive definite matrix containing inertia parameters I_x, I_y and I_z ; $F_{ext} \in \mathbb{R}^3$ and $\tau_{ext} \in \mathbb{R}^3$ are the external forces and torques, respectively, $C(\eta, \dot{\eta})$ is the Coriolis term given in [2], and Ψ_η represents Euler parameter given by

$$\Psi_\eta = \begin{pmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \phi & \sin \phi \cos \theta \\ 0 & -\sin \phi & \cos \phi \cos \theta \end{pmatrix} \quad (3)$$

The two rotors generate the thrust T whose expression can be defined through the rotor angular speeds Ω_1 and Ω_2 , and the

swash-plate incidence angles δ_{cx} and δ_{cy} as

$$T = \begin{pmatrix} T_x \\ T_y \\ T_z \end{pmatrix} = \begin{pmatrix} -k_\beta \sin \delta_{cy} \cos \delta_{cx} \Omega_2^2 \\ -k_\beta \sin \delta_{cx} \Omega_2^2 \\ k_\alpha \Omega_1^2 + k_\beta \cos \delta_{cx} \cos \delta_{cy} \Omega_2^2 \end{pmatrix} \quad (4)$$

where k_α and k_β are positive rotor aerodynamic coefficients. The separation between the center of the mass and the rotor hubs generates a torque vector τ defined by

$$\tau = \begin{pmatrix} \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{pmatrix} = \begin{pmatrix} -dk_\beta \sin \delta_{cx} \Omega_2^2 \\ dk_\beta \sin \delta_{cy} \cos \delta_{cx} \Omega_2^2 \\ \gamma_1 \Omega_1^2 - \gamma_2 \Omega_2^2 \end{pmatrix} \quad (5)$$

where γ_1 and γ_2 represent the positive yaw aerodynamic coefficients and d is the distance between the center of gravity and the lower rotor center of rotation.

Remark 1. From a practical point of view, we can simplify the expression model by considering that δ_{cx} , δ_{cy} are small, which leads to $\cos \delta_{cx} \approx \cos \delta_{cy} \approx 1$, $\sin \delta_{cx} \approx \delta_{cx}$ and $\sin \delta_{cy} \approx \delta_{cy}$ [10]. Therefore, the lateral action caused by T_x and T_y in (4) can be assumed negligible compared to the contribution of main thrust T_z .

By using Eqs. (2), (4), (5) and considering Remark 1, we can get the coaxial rotor model in Earth-frame $E(x_e, y_e, z_e)$ given by [12]

$$\begin{cases} \ddot{x} = \frac{(\cos \phi \sin \theta \cos \psi + \sin \phi \sin \psi)(k_\alpha \Omega_1^2 + k_\beta \cos \delta_{cx} \cos \delta_{cy} \Omega_2^2)}{m} \\ \ddot{y} = \frac{(\cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi)(k_\alpha \Omega_1^2 + k_\beta \cos \delta_{cx} \cos \delta_{cy} \Omega_2^2)}{m} \\ \ddot{z} = \frac{(\cos \phi \cos \theta)(k_\alpha \Omega_1^2 + k_\beta \cos \delta_{cx} \cos \delta_{cy} \Omega_2^2)}{m} - g \\ \ddot{\phi} = \frac{l_y - l_z}{I_x} \dot{\theta} \dot{\psi} - \frac{dk_\beta \sin \delta_{cx} \Omega_2^2}{I_x} \\ \ddot{\theta} = \frac{l_z - l_x}{I_y} \dot{\phi} \dot{\psi} + \frac{dk_\beta \sin \delta_{cy} \cos \delta_{cx} \Omega_2^2}{I_y} \\ \ddot{\psi} = \frac{l_x - l_y}{I_z} \dot{\phi} \dot{\theta} + \frac{\gamma_1 \Omega_1^2 - \gamma_2 \Omega_2^2}{I_z} \end{cases} \quad (6)$$

It is important to note that the position and attitude of the coaxial rotor obtained in the final model (6) are free from external disturbances effect ($F_{ext} = 0$ and $\tau_{ext} = 0$), and they are controlled through the input signals Ω_1 , Ω_2 , δ_{cx} , δ_{cy} , which can be obtained by applying the inverse transformation to Eqs. (4) and (5). Their expressions are given by

$$\begin{cases} \Omega_1^2 = \frac{\gamma_2 T_z - k_\beta \tau_\psi}{k_\alpha \gamma_2 + k_\beta \gamma_1}, \\ \Omega_2^2 = \frac{\gamma_1 T_z - k_\alpha \tau_\psi}{k_\alpha \gamma_2 + k_\beta \gamma_1}, \\ \delta_{cx} = -\frac{\tau_\phi}{dk_\beta \Omega_2^2}, \\ \delta_{cy} = \frac{\tau_\theta}{dk_\beta \Omega_2^2}. \end{cases} \quad (7)$$

Remark 2. To ensure system controllability, Euler angles (roll ϕ , pitch θ and yaw ψ) must respect the conditions $|\phi| < \frac{\pi}{2}$, $|\theta| < \frac{\pi}{2}$ and $|\psi| < \pi$.

2.2. Problem statement

From the equation of motion (6), it is clear that the coaxial rotor model consists of two groups of nonlinear subsystems in series and each group comprises three subsystems. The first group of subsystems, which is an underactuated system as it is only controlled by the force T_z , describes the coaxial position dynamics along the x , y and z axes. The second group of subsystems comprises the three subsystems describing the orientation through the roll ϕ , pitch θ and yaw ψ angles controlled by the three torques τ_ϕ , τ_θ and τ_ψ , respectively. Therefore, the coaxial

dynamic model can be represented by six subsystems $i \in \mathcal{E} = \{x, y, z, \phi, \theta, \psi\}$, each one driven via a control input.

The position is controlled through three virtual inputs u_x , u_y and u_z introduced to deal with the underactuated nature of coaxial rotor. Their expressions are chosen as follows:

$$(u_x, u_y, u_z)^T = T_z R(0, 0, 1)^T - g(0, 0, 1)^T \quad (8)$$

The coaxial position and orientation represent the output of the system $y_i \in \mathcal{E}$ that must be controlled, while the desired flight trajectory is translated in the desired position and the yaw angle respectively x_d , y_d , z_d , and ψ_d which are the inputs of the system. The desired pitch and roll angles ϕ_d , θ_d and the desired force vector magnitude T_z are obtained from the position subsystems group according to Eq. (9) as follows:

$$\begin{pmatrix} T_z \\ \phi_d \\ \theta_d \end{pmatrix} = \begin{pmatrix} \sqrt{u_x^2 + u_y^2 + (u_z + g)^2} \\ \sin^{-1}[\frac{1}{T_z}(u_x \sin \psi_d - u_y \cos \psi_d)] \\ \sin^{-1}[\frac{1}{T_z \cos \phi_d}(u_x \cos \psi_d + u_y \sin \psi_d)] \end{pmatrix} \quad (9)$$

Taking into account transformations (8) and (9), the dynamic model (6) of the coaxial system can be represented in state-space form for the six interconnected subsystems as follows:

$$\begin{cases} \dot{x}_{i1} = x_{i2}, \\ \dot{x}_{i2} = f_i(\mathcal{X}) + b_i u_i(t) + h_i(t), i \in \mathcal{E} \\ y_i = x_{i1} \end{cases} \quad (10)$$

where $\mathcal{X} = [x_{x1}, x_{x2}, \dots, x_{\psi 1}, x_{\psi 2}]^T \in \mathbb{R}^{12}$ represents the total state vector of the coaxial rotor, $x_i = [x_{i1}, x_{i2}]^T \in \mathbb{R}^2$ denotes the state vector of each subsystem and $u_i \in \{u_x, u_y, u_z, u_\phi, u_\theta, u_\psi\}$ its corresponding control input, $h_i(t)$ are bounded external disturbances, and $f_i(\mathcal{X})$ and b_i are nonlinear dynamic functions and the control input gains, respectively, with

$$f_x(\mathcal{X}) = 0, f_y(\mathcal{X}) = 0, f_z(\mathcal{X}) = 0, f_\phi(\mathcal{X}) = \frac{l_y - l_z}{I_x} x_{\theta 2} x_{\psi 2},$$

$$f_\theta(\mathcal{X}) = \frac{l_z - l_x}{I_y} x_{\phi 2} x_{\psi 2},$$

$$f_\psi(\mathcal{X}) = \frac{l_x - l_y}{I_z} x_{\phi 2} x_{\theta 2},$$

$$b_x = 1/m, b_y = 1/m, b_z = 1/m, b_\phi = d/I_x, b_\theta = d/I_y, b_\psi = 1/I_z.$$

To facilitate our control design, we combined the dynamic functions and external perturbation of each coaxial rotor subsystem in one nonlinear function to be estimated. Multiplying the second equation of (10) on the left by b_i^{-1} and adding $\dot{x}_{i2} - b_i^{-1} \dot{x}_{i2}$ to both sides, we can rewrite it as

$$\begin{cases} \dot{x}_{i1} = x_{i2}, \\ \dot{x}_{i2} = \Psi_i(\mathcal{X}, t) + u_i(t), i \in \mathcal{E} \\ y_i = x_{i1}. \end{cases} \quad (11)$$

where $\Psi_i(\mathcal{X}, t) = (1 - b_i^{-1})\dot{x}_{i2} + b_i^{-1}f_i(\mathcal{X}) + b_i^{-1}h_i(t)$ represents the entire of nonlinear functionals with the disturbance of each subsystem.

Considering the transformations in Eq. (11), a simplified dynamic model was derived to obtain a controller that is easy and simple to implement. **This controller can also effectively deal with unknown dynamics and disturbances grouped in $\Psi_i(\mathcal{X}, t)$.** Thus, given the smoothness, continuous, and boundness of the desired trajectory signals $y_{id} \in \{x_d, y_d, z_d, \phi_d, \theta_d, \psi_d\}$, the objective is to design a robust control u_i with an optimal model-free scheme such that the position and attitude y_i of the coaxial rotor track their desired paths y_{id} with a fast convergence time, null steady-state error, and insensitivity to the unknown dynamics and external disturbances.

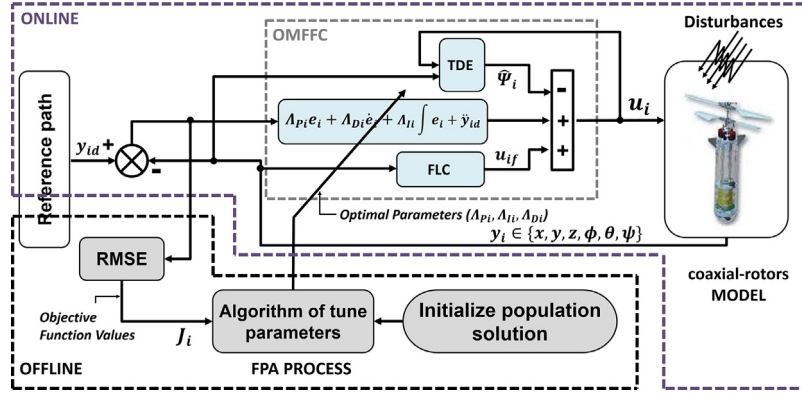


Fig. 2. Optimal model-free fuzzy control (OMFFC) structure.

3. Control design

This part provides a description of the proposed OMFFC for position and attitude control of a coaxial rotor drone. We propose here a novel control strategy based on an optimal proportional–integral–derivative (PID) controller and the time-delay estimation (TDE) technique to deal with the unknown dynamic functions of the system and disturbances. To offset the TDE estimation error, a robust adaptive fuzzy logic compensator is then introduced. Finally, the optimal parameters of the design conditions guaranteeing optimal tracking performance are selected using a flower pollination-based optimization algorithm. Some details of the design of the closed-loop control structure (shown in Fig. 2) are discussed in the following.

3.1. Time-delay estimation control

In this section, the design process for a model-free time-delay compensation controller is developed for the coaxial rotor system formulated in (11), as in [32]. The nonlinear dynamics and disturbance of each subsystem of the coaxial device, grouped in $\Psi_i(\mathcal{X}, t)$, are approximated using the time delay estimation (TDE) technique to provide a simple and efficient control scheme. To elaborate our control strategy, let us define the tracking errors e_i as follows:

$$e_i = y_{id} - y_i \quad (12)$$

Their derivatives with respect to time $\dot{e}_i$ and $\ddot{e}_i$ are

$$\dot{e}_i = \dot{y}_{id} - \dot{y}_i = \dot{y}_{id} - \dot{x}_{i1} \quad (13)$$

$$\ddot{e}_i = \ddot{y}_{id} - \ddot{y}_i = \ddot{y}_{id} - \dot{x}_{i2} \quad (14)$$

where $\dot{y}_{id}$ and $\ddot{y}_{id}$ represent first and second order time derivative of y_{id} , respectively. Their values are assumed to be known and bounded. Substituting Eqs. (12) to (14) into Eq. (11), the coaxial rotor dynamics can be described by the tracking error equations as

$$\ddot{e}_i - \ddot{y}_{id} + \Psi_i(\mathcal{X}, t) + u_i(t) = 0 \quad (15)$$

From Eq. (15), the tracking errors could converge to zero if $\Psi_i(\mathcal{X}, t)$ are well known and the control law can be dedicated directly by using a compensation technique such as $u_i(t) = \ddot{y}_{id} + u_{li} - \Psi_i(\mathcal{X}, t)$, where u_{li} are linear control terms chosen to have pre-assigned performances for the coaxial system. However, the information in $\Psi_i(\mathcal{X}, t)$ can only be obtained if the coaxial motion dynamic model is known accurately and is free of any parametric variations and external disturbances. To overcome this problem, the TDE approach is employed in combination with

a linear proportional–derivative (PD) controller to approximate this ideal control law in the following form:

$$u_i(t) = \ddot{y}_{id} + \Gamma_{Pi} e_i + \Gamma_{Di} \dot{e}_i - \hat{\Psi}_i(\mathcal{X}, t) \quad (16)$$

where $\hat{\Psi}_i(\mathcal{X}, t)$ are the approximate values of $\Psi_i(\mathcal{X}, t)$. $\Gamma_{Pi} > 0$ and $\Gamma_{Di} > 0$ are controller gains selected by the designer to have a stable closed-loop system according to the following error equation:

$$\Gamma_{Pi} e_i + \Gamma_{Di} \dot{e}_i + \ddot{e}_i = 0 \quad (17)$$

The tracking performance of the coaxial system is maintained by setting Γ_{Pi} and Γ_{Di} using the pole placement method to obtain good tracking of the desired reference signal. The estimators $\hat{\Psi}_i(\mathcal{X}, t)$ take their expressions according to the TDE formulation [32] by considering the state model (11) which leads to the following form:

$$\hat{\Psi}_i(\mathcal{X}, t) = \Psi_i(\mathcal{X}, t - \rho) = u_i(t - \rho) - \ddot{y}_i(t - \rho) \quad (18)$$

where ρ is the time-delay estimation constant generally set as one sampling period. By substituting (18) in (16), we can obtain the final time-delay estimation control (TDEC) law:

$$u_i(t) = \ddot{y}_{id} + \Gamma_{Pi} e_i + \Gamma_{Di} \dot{e}_i - u_i(t - \rho) + \ddot{y}_i(t - \rho) \quad (19)$$

As in [32,34], the TDEC algorithm (19) can effectively stabilize the given drone system. Therefore, the controller design becomes straightforward and no dynamics are required to handle trajectory tracking.

Remark 3. The nonlinear functions $\Psi_i(\mathcal{X}, t)$ are supposed to be continuous and vary slowly with respect to time so that they can be estimated simply and accurately by the TDE approach in a short sampling period ρ . The outputs of coaxial rotor y_i are assumed to be available for measurement and the derivatives $\dot{y}_i$ and $\ddot{y}_i$ are calculated after filtering using mathematical differentiation.

3.2. Model-free fuzzy logic controller

The TDEC scheme developed in the previous subsection is the basic model-free control structure proposed to drive the coaxial rotor system. This controller is obtained with unrealistic assumptions that the coaxial rotor dynamics and disturbances grouped in $\Psi_i(\mathcal{X}, t)$ are ideally approximated by the TDE technique (18) with neglecting the estimation error in (17). Besides, it cannot achieve a null static error using a PD control term. To tackle these disadvantages, a robust adaptive fuzzy compensator is added to the time-delay estimation with a linear PID controller for which the desired control law is designed as

$$u_i(t) = \ddot{y}_{id} + \Lambda_{Pi} e_i + \Lambda_{Di} \dot{e}_i + \Lambda_{Ii} \int e_i - \hat{\Psi}_i(\mathcal{X}, t) + \tau_{fi} \quad (20)$$

where Λ_{pi} , Λ_{Di} and Λ_{fi} are the PID tuning parameters and τ_{fi} is the local fuzzy logic control term of each subsystem, i.e. τ_{fx} , τ_{fy} and τ_{fz} are fuzzy controls for position x , y and z subsystems, respectively when $\tau_{f\phi}$, $\tau_{f\theta}$ and $\tau_{f\psi}$ are fuzzy control inputs for attitude ϕ , θ and ψ subsystems. The estimators $\tilde{\Psi}_i(\chi, t)$ are calculated similarly according to Eq. (18).

From Eqs. (18) and (20), it can be found that the designed control law for position and attitude subsystems is related only to the local input/output data measurements, u_i and y_i , without needing dynamic model information. Therefore, the adopted controller in this paper is free to any model of coaxial rotor with a simple decentralized structure, making the given control algorithm easier to be implemented unlike to other MFC controllers [26,28,30].

Introducing control law (20) into Eq. (15) yields the error dynamic equation of the closed-loop system as follows

$$\ddot{e}_i + \Lambda_{Di}\dot{e}_i + \Lambda_{pi}e_i + \Lambda_{fi} \int e_i = \tilde{\Psi}_i(\chi, t) + \tau_{fi} \quad (21)$$

where $\tilde{\Psi}_i(\chi, t) = \Psi_i(\chi, t) - \hat{\Psi}_i(\chi, t)$ are the estimation errors. Eq. (21) is similar to Eq. (17) when $\tilde{\Psi}_i(\chi, t) \rightarrow 0$ for $t \rightarrow \infty$ with a small estimation error. Thereby, the error dynamic of the coaxial control system is linked to the appropriate selection of the design parameters Λ_{pi} , Λ_{fi} and Λ_{Di} , their values are selected so that the tracking errors e_i converge to zero and the closed-loop stability is guaranteed. Regarding TDE-errors have values that are significant and not neglected, i.e. $\tilde{\Psi}_i(\chi, t) \neq 0$ for $t \rightarrow \infty$, the additional control terms τ_{fi} are designed to offset them using an adaptive fuzzy inference system because of its simplicity and universal approximation property.

For each subsystem, let put $\mathcal{E}_i = [e_i, \dot{e}_i]^T$, then (21) becomes

$$\dot{\mathcal{E}}_i = A_i \mathcal{E}_i + B_i (\tilde{\Theta}_i(\chi, t) + \tau_{fi}) \quad (22)$$

With $A_i = \begin{bmatrix} 0 & 1 \\ -\Lambda_{pi} & -\Lambda_{Di} \end{bmatrix}$, $B_i = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\tilde{\Theta}_i(\chi, t) = \tilde{\Psi}_i(\chi, t) - k_{il} \int e_i$

If $\tau_{fi} \rightarrow -\tilde{\Theta}_i(\chi, t)$ the coaxial motion controller makes it possible to obtain a substantial improvement of system performances. Using Mamdani fuzzy system for the input vector $\mathcal{E}_i$, the ideal values of $\tilde{\Theta}_i(\chi, t)$ can be approximated as follows [38]

$$\tilde{\Theta}_i^*(\chi, t) = \varphi_i^{*T} \xi_i(e_i, \dot{e}_i) + \Delta_i \quad (23)$$

where Δ_i represent the compensation fuzzy errors satisfying, according to the universal approximation theory [39], the following inequality

$$|\Delta_i(t)| < \bar{\Delta}_i \quad (24)$$

with $\bar{\Delta}_i$ are unknown positive constants.

$\xi_i(e_i, \dot{e}_i) = [\xi_{i,1}(e_i, \dot{e}_i), \dots, \xi_{i,N}(e_i, \dot{e}_i)]^T$ are basic fuzzy function vectors of N the fuzzy rules “If-Then”. Their elements are calculated from selected membership functions $\mu_{\check{e}_i^h}^h(e_i)$ and $\mu_{\check{e}_i^h}^h(\dot{e}_i)$, by applying product-inference and weighted average methods according to the following relationship

$$\xi_{i,h}(e_i) = \frac{\mu_{\check{e}_i^h}^h(e_{pi}) \mu_{\sigma_{vi}^h}(e_{vi})}{\sum_{k=1}^N \mu_{\check{e}_i^k}^h(e_{pi}) \mu_{\sigma_{vi}^k}^h(e_{vi})} h = 1, \dots, N \quad (25)$$

where $\check{e}_i^h \in \{\check{e}_{e_i}^1, \dots, \check{e}_{e_i}^{m_{e_i}}\}$ and $\check{e}_i^h \in \{\check{e}_{\dot{e}_i}^1, \dots, \check{e}_{\dot{e}_i}^{m_{\dot{e}_i}}\}$ are the fuzzy sets of e_i and $\dot{e}_i$ respectively, and $\varphi_i^* = [\varphi_{i,1}^*, \dots, \varphi_{i,N}^*]^T$ are the ideal parameters minimizing the approximation errors Δ_i that satisfy:

$$\varphi_i^* = \underset{\varphi_i}{\operatorname{argmin}} \left[\sup_{e_i \in \mathcal{N}_{e_i}} |\tilde{\Theta}_i(\chi, t) - \tilde{\Theta}_i^*(e_i, \dot{e}_i, \varphi_i^*)| \right]$$

Then, Eq. (22) becomes

$$\mathcal{E}_i = A_i \mathcal{E}_i + B_i (\varphi_i^{*T} \xi_i(e_i, \dot{e}_i) + \Delta_i + \tau_{fi}) \quad (26)$$

To make the tracking errors converge to zero, i.e. $e_i(t) \rightarrow 0$ for $t \rightarrow 0$, and compensate the estimation errors, the fuzzy control terms are designed as

$$\tau_{fi} = -\hat{\Theta}_i(e_i, \dot{e}_i, \hat{\varphi}_i) = -\hat{\varphi}_i^T \xi_i(e_i, \dot{e}_i) - \hat{\Delta}_i \operatorname{sgn}(\mathcal{E}_i^T P_i B_i) \quad (27)$$

where $\hat{\varphi}_i$ and $\hat{\Delta}_i$ are respectively the estimates of φ_i^* and $\bar{\Delta}_i$, and $\operatorname{sgn}(\cdot)$ denotes the sign function replaced in simulations by $\tanh(\cdot)$ to avoid chattering problem. The computational cost of τ_{fi} is reduced by considering a minimum number of fuzzy sets (three fuzzy sets N , Z and P) for every input e_i and $\dot{e}_i$ where each subsystem has one fuzzy system to compensate the corresponding TDE-error.

The stability of the overall coaxial rotor control system is investigated by using the Lyapunov method through the following theorem.

Theorem. Consider the dynamics of coaxial rotor UAV described as (11) with the proposed MFFC law in (20), TDE-based control term in (18), and the fuzzy compensator in (27). The position and attitude tracking errors e_i of the coaxial rotor converge to zero and the closed-loop system can achieve asymptotic stability if the following adaptive laws hold

$$\dot{\hat{\varphi}}_i = \vartheta_i \mathcal{E}_i^T P_i B_i \xi_i(e_i, \dot{e}_i) \quad (28)$$

$$\dot{\hat{\Delta}}_i = \eta_i |\mathcal{E}_i^T P_i B_i| \quad (29)$$

where ϑ_i and η_i are strictly positive design parameters, and $P_i > 0$ are symmetric matrices solution of the following equation

$$P_i A_i^T + P_i A_i = -Q_i \quad (30)$$

with $Q_i > 0$ are diagonal matrices chosen by the user.

Proof. Let us consider the Lyapunov function as:

$$V = \sum_{i \in \mathcal{S}} V_i \quad (31)$$

where V_i is the subsystem stability function, with

$$V_i = \frac{1}{2} \mathcal{E}_i^T P_i \mathcal{E}_i + \frac{1}{2\vartheta_i} \tilde{\varphi}_i^T \tilde{\varphi}_i + \frac{1}{2\eta_i} \tilde{\Delta}_i^2 \quad (32)$$

where $\tilde{\varphi}_i = \varphi_i^* - \hat{\varphi}_i$ and $\tilde{\Delta}_i = \bar{\Delta}_i - \hat{\Delta}_i$ are the estimation errors.

The differentiation of (31) along (26), gives

$$\dot{V} = \sum_{i \in \mathcal{S}} \dot{V}_i = \sum_{i \in \mathcal{S}} \frac{1}{2} \left(\dot{\mathcal{E}}_i^T P_i \mathcal{E}_i + \mathcal{E}_i^T P_i \dot{\mathcal{E}}_i \right) - \frac{1}{\vartheta_i} \tilde{\varphi}_i^T \dot{\tilde{\varphi}}_i - \frac{1}{\eta_i} \tilde{\Delta}_i \dot{\tilde{\Delta}}_i \quad (33)$$

By substituting (26) and (27) into (33), $\dot{V}$ becomes

$$\begin{aligned} \dot{V} &= \sum_{i \in \mathcal{S}} \frac{1}{2} \left(A_i \mathcal{E}_i + B_i \left(\tilde{\varphi}_i^T \xi_i(e_i, \dot{e}_i) + \Delta_i - \hat{\Delta}_i \operatorname{sgn}(\mathcal{E}_i^T P_i B_i) \right) \right)^T P_i \mathcal{E}_i \\ &\quad + \frac{1}{2} \mathcal{E}_i^T P_i \left(A_i \mathcal{E}_i + B_i \left(\tilde{\varphi}_i^T \xi_i(e_i, \dot{e}_i) + \Delta_i - \hat{\Delta}_i \operatorname{sgn}(\mathcal{E}_i^T P_i B_i) \right) \right) \\ &\quad - \frac{1}{\vartheta_i} \tilde{\varphi}_i^T \dot{\tilde{\varphi}}_i - \frac{1}{\eta_i} \tilde{\Delta}_i \dot{\tilde{\Delta}}_i \\ &= \sum_{i \in \mathcal{S}} \frac{1}{2} \mathcal{E}_i^T \left(P_i A_i + A_i^T P_i \right) \mathcal{E}_i + \tilde{\varphi}_i^T \left(\mathcal{E}_i^T P_i B_i \xi_i(e_i, \dot{e}_i) - \frac{1}{\vartheta_i} \dot{\tilde{\varphi}}_i \right) \\ &\quad + \mathcal{E}_i^T P_i B_i \Delta_i - \hat{\Delta}_i \mathcal{E}_i^T P_i B_i \operatorname{sgn}(\mathcal{E}_i^T P_i B_i) - \frac{1}{\eta_i} \tilde{\Delta}_i \dot{\tilde{\Delta}}_i \end{aligned} \quad (34)$$

Using the adaptive law (28) with (30), yields

$$\dot{V} = \sum_{i \in \mathcal{S}} -\frac{1}{2} \mathcal{E}_i^T Q_i \mathcal{E}_i + \mathcal{E}_i^T P_i B_i \Delta_i - \hat{\Delta}_i \mathcal{E}_i^T P_i B_i \operatorname{sgn}(\mathcal{E}_i^T P_i B_i) - \frac{1}{\eta_i} \tilde{\Delta}_i \dot{\tilde{\Delta}}_i \quad (35)$$

According to Cauchy–Schwarz inequality and using (24), we can easily obtain

$$\begin{aligned} \dot{V} &\leq \sum_{i \in \mathcal{E}} -\frac{1}{2} \varepsilon_i^T Q_i \varepsilon_i + |\varepsilon_i^T P_i B_i| |\Delta_i| - \hat{\Delta}_i |\varepsilon_i^T P_i B_i| - \frac{1}{\eta_i} \tilde{\Delta}_i \dot{\Delta}_i \\ &\leq \sum_{i \in \mathcal{E}} -\frac{1}{2} \varepsilon_i^T Q_i \varepsilon_i + |\varepsilon_i^T P_i B_i| \tilde{\Delta}_i - \hat{\Delta}_i |\varepsilon_i^T P_i B_i| - \frac{1}{\eta_i} \tilde{\Delta}_i \dot{\Delta}_i \end{aligned} \quad (36)$$

Finally, by applying (29), we can get

$$\begin{aligned} \dot{V} &\leq \sum_{i \in \mathcal{E}} -\frac{1}{2} \varepsilon_i^T Q_i \varepsilon_i + |\varepsilon_i^T P_i B_i| \tilde{\Delta}_i - \frac{1}{\eta_i} \tilde{\Delta}_i (\eta_i |\varepsilon_i^T P_i B_i|) \\ &\leq \sum_{i \in \mathcal{E}} -\frac{1}{2} \varepsilon_i^T Q_i \varepsilon_i \leq \sum_{i \in \mathcal{E}} -\frac{1}{2} \varrho_{i,\min}(Q_i) |\varepsilon_i|^2 \end{aligned} \quad (37)$$

where $\varrho_{i,\min}(Q_i)$ are the minimum eigenvalues of the matrix Q_i . From (31) to (37), it leads to $V \in L_\infty$ meaning that the given drone system is globally asymptotically stable using the designed MFFC in Eq. (20) and the variables e_i , $\dot{e}_i$, $\tilde{\varphi}_i$, $\tilde{\Delta}_i$ and u_i are bounded. Furthermore, e_i and $\dot{e}_i$ converge asymptotically to zero when $t \rightarrow +\infty$ according to Barbalat's lemma [40].

Remark 4. The coaxial rotor system, in Eq. (11), ensures the stability property by applying the MFFC law in (20). The MFFC law depends only on the tracking errors and their first derivatives that are e_i and $\dot{e}_i$. The outputs y_i , which are the position and the attitude, achieve asymptotic convergence to the desired given reference y_{id} by utilizing the updating laws (28) and (29) derived from the stability analysis using the Lyapunov method. There are three parameters to be designed, Λ_{Pi} , Λ_{Di} and Λ_{Ii} , for each coaxial rotor subsystem. These parameters can be determined based on the requirements of the steady-state precision or the rapidity of the transient dynamics and the adaptation speed of the control signal. In this paper, they are tuned offline by means of a metaheuristic optimization method in order to achieve the best performance of the drone. This design phase will be expanded in the next section.

3.3. Optimal model-free fuzzy logic control law's design

This part of our control strategy is devoted to tuning the parameters' value of the MFFC control law using a bio-inspired metaheuristic optimization algorithm, namely the flower pollination algorithm (FPA). The FPA is a metaheuristic algorithm

introduced by Xin-She Yang [36] inspired by biological flower pollination in flowering plants. This algorithm is based on the following rules:

- Global pollination processes can be achieved using pollen of the same flower and using pollen from a flower of some other plants of the same species (the pollen-carrying pollinators move in a way that obeys the Levy flight).
- Abiotic and self-pollination occur for local pollination without the help of any insects.
- Flower constancy can be developed with the help of pollinators such as insects. Reproduction probability constancy is proportional to the similarity of the two flowers involved.
- A switch probability $\wp$ can control the interaction of local pollination and global pollination, slightly biased toward local pollination because of physical proximity and wind.

Let the solution of the t th iteration of flower j be S_j^t and the best flower of the global population vector be S^* . The iteration flower constancy can be represented as

$$S_j^{t+1} = S_j^t + \gamma L(\lambda)(S^* - S_j^t) \quad (38)$$

where γ is constant, and $L(\lambda)$ is the strength of the pollination (the step size of the Levy flight). The Levy distribution is used to represent the flight of the insects as in

$$L \sim \frac{\lambda v(\lambda) \sin(\varphi\lambda/2)}{\pi} \frac{1}{l^{1+\lambda}} \quad (39)$$

where $\pi \approx 3.14$, λ is constant, $v(\lambda)$ represents the standard gamma function, $l = \frac{U}{|W|^{1/\lambda}}$, $U \sim N(0, \sigma^2)$, $W \sim N(0, 1)$ with $\sigma^2 = (\frac{v(1+\lambda)}{\lambda v((1+\lambda)/2)} \cdot \frac{\sin(\pi\lambda/2)}{2^{(\lambda-1)/2}})^{1/\lambda}$ [36]. In some flowers, self-pollination takes place with the following dynamics

$$v_j^{t+1} = S_j^t + \varepsilon(S_n^t - S_k^t) \quad (40)$$

where $\varepsilon \in [0, 1]$ is a random number, S_n^t, S_k^t are the pollen from different flower positions in the population. For each flower, the interaction or switching from common global pollination to intensive local pollination can be controlled using the switching probability $\wp$ as

$$S_j^{t+1} = \begin{cases} v_j^{t+1} & \text{if rand}(0, 1) > \wp \\ S_j^{t+1} & \text{Otherwise} \end{cases} \quad (41)$$

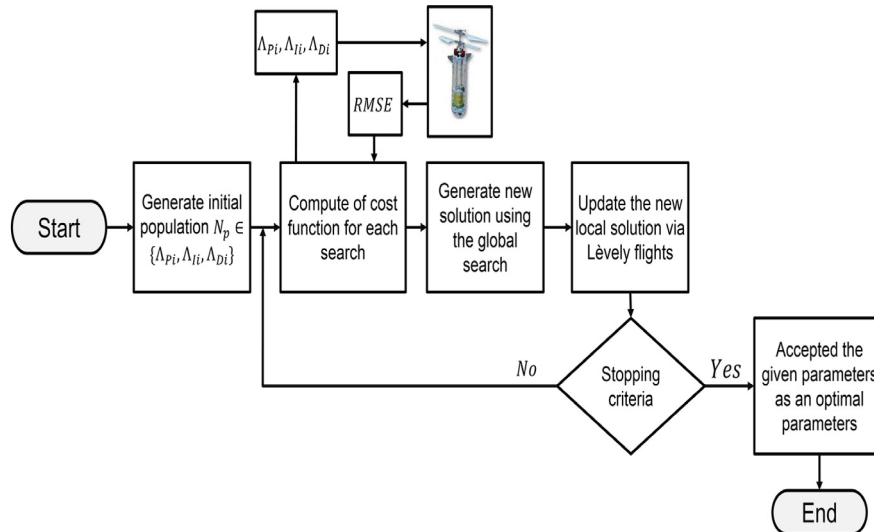


Fig. 3. Flowchart of FPA process.

Table 1
Coaxial rotor parameters.

Parameters	Value
I_x	$1.383 \times 10^{-3} \text{ kg m}^2$
I_y	$1.383 \times 10^{-3} \text{ kg m}^2$
I_z	$2.72 \times 10^{-4} \text{ kg m}^2$
k_α	$3.683 \times 10^{-5} \text{ N/rad}^2 \text{ s}^2$
k_β	$3.776 \times 10^{-5} \text{ N/rad}^2 \text{ s}^2$
γ_1	$1.476 \times 10^{-6} \text{ N m/rad}^2 \text{ s}^2$
γ_2	$1.326 \times 10^{-6} \text{ N m/rad}^2 \text{ s}^2$
g	9.81 m/s^2
d	0.0676 m
m	0.41 kg

The selection operator of the best current global flower depends on the flower that has the lowest value of the objective function during pollination.

In the following, we propose the FPA to compute the optimal value of parameters Λ_{pi} , Λ_{ji} and Λ_{Di} for the control law (20) minimizing an objective function J_i . The step-by-step procedure of the FPA to tune optimally the parameters' value of the designed controller is given below.

Step 1: Set the population size N_p , the maximum number of iterations, the switching probability $\wp$, the scaling factor ε , the step size j , and the constant γ .

Step 2: Generate the initial feasible solutions according to the Lower-bound S_L and the Upper-bound S_U , that are the same size as the dimensional problem. Compute the root mean square error (RMSE) of each initial solution, and identify the initial global best solution S^* , $\forall S_j^t = (\Lambda_{pi}, \Lambda_{ji}, \Lambda_{Di})$. The RMSE criterion can be defined as

$$J_i = \sqrt{\frac{\sum_{\ell=1}^N (y_{id} - y_i)^2}{N}} \quad (42)$$

with N is the sampling time size.

In this work, the optimization objective functions for the coaxial rotor are chosen as

$$J_p = \frac{\sum_{i \in \{x, y, z\}} J_i}{3} \text{ and } J_R = \frac{\sum_{i \in \{\phi, \theta, \psi\}} J_i}{3}.$$

where J_p and J_R are the RMSE's of the position and attitude controllers, respectively.

Step 3: Use the switching probability (41). If $\text{rand} < p$, use global pollination (38); else, use local pollination (40). If any component of the solution violates the limits, generate new initial feasible solutions.

Step 4: Evaluate new solutions. Replace the old solutions with the new best solutions. The best solutions of flowers are chosen in such a way as to minimize the objective function. The previous steps should be repeated until the stopping criterion is reached. The flower pollination algorithm is summarized in the flowchart depicted in Fig. 3. Finally, the control of the coaxial rotor is performed using the developed OMFFC through two cascade control loops (outer and inner), each one with different focus. The outer loop is designed for the position tracking, whilst the inner loop is designed to stabilize the attitude.

4. Numerical experiments

In this section, the performance of the designed optimal model-free fuzzy controller (OMFFC) is evaluated through various simulation cases. The coaxial rotor model, described in Section 2, is used to validate numerically the proposed MFC control approach. This mathematical model, obtained from several works in the

Table 2
Parameters for the FPA.

Parameters	Value
Probability switch	0.8
γ	1.5
Scaling factor	0.1
Iteration	100

literature, is already involved for numerical and experimental validation purposes [3,10]. The physical parameters of the coaxial rotor are shown in Table 1 [10]. The initial conditions for the coaxial rotor are $P_0 = [0, 0, 0]^T \text{ m}$ and $\eta_0 = [0, 0, 0]^T \text{ rad}$ for all the simulations.

The proposed controller was optimized in a bounded search space $S \in [S_L^i, S_U^i] = [1, 30]$, according to the steps mentioned in the previous section. The objective was to obtain the optimum design parameters to be used in the simulation tests. The tuning parameter values of the FPA are listed in Table 2, whilst Table 3 gives the parameters' values of the designed controller in (20), the PID and the TDEC control algorithms. To show high performances of the coaxial rotor using the developed control method, three simulation tests are presented: optimization performance, robustness comparison study with other related controllers, and sensitivity to sensor noise and parametric variations.

4.1. Test 1: Optimization performance

For the primary test, the OMFFC proposed was compared with two MFFCs without optimization. The parameters, set 1 and set 2, were chosen arbitrarily using a trial-and-error method. The desired path, in this case, was chosen as a square form for all the coaxial rotor subsystems defined by the following equations:

$$x_d = \begin{cases} 2 \text{ m}, & t < 20 \text{ s} \\ 0 \text{ m}, & 20 \leq t < 35 \text{ s} \\ 2 \text{ m}, & 35 \leq t \end{cases}, \quad y_d = \begin{cases} 0 \text{ m}, & t < 10 \text{ s} \\ 2 \text{ m}, & 10 \leq t < 30 \text{ s} \\ 1 \text{ m}, & 30 \leq t \end{cases},$$

$$z_d = 3 \text{ m and } \psi_d = 1 \text{ rad.}$$

The results obtained are depicted in Figs. 4–6. Fig. 4 depicts the tracking response of the position subsystems and shows that the coaxial rotor can perfectly track the planned trajectory with the smallest error when an optimization process is applied. The proposed OMFFC presents considerably better tracking performances than the other controllers with design parameters set 1 and set 2. The attitude roll, pitch, and yaw angle responses are depicted in Fig. 5. From the step response of the yaw (ψ) angle, the proposed method (blue solid line in Fig. 5) achieved the best tracking accuracy, overshoot, and tracking speed, which clearly shows the effectiveness of the FPA optimization. The results confirm that the designed OMFFC exhibits better performance than the other set 1 and set 2 controllers. The behavior of the control inputs displayed in Fig. 6 is smooth without coupling effect and could easily be implemented in a practical system.

4.2. Test 2: Comparison study

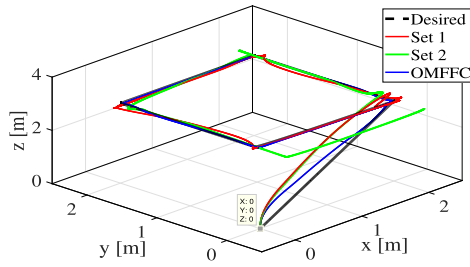
This test evaluated the performance of the proposed OMFFC under disturbances compared to two related controllers: the PID and TDEC, in a circular path defined as $x_d = 2 \sin 0.2t \text{ m}$, $y_d = 2 \cos 0.2t \text{ m}$, $z_d = 3 \text{ m}$ and $\psi_d = \begin{cases} 0.5 \text{ rad}, & t < 20 \text{ s} \\ -0.5 \text{ rad}, & 20 \leq t \end{cases}$.

The same sinusoidal disturbances $h_i(t)$ for $i \in \{x, y, z\}$ used in [10] were proposed in this test at 20, 30 and 10 s, respectively

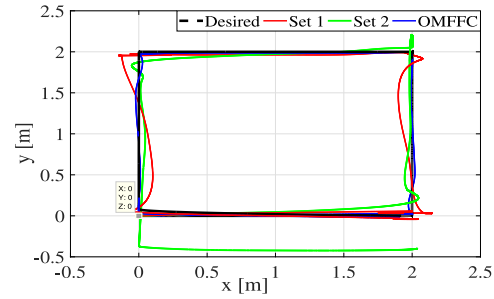
$$\begin{pmatrix} h_x(t) \\ h_y(t) \\ h_z(t) \end{pmatrix} = \begin{pmatrix} \sin 0.1t \\ \sin 0.1t \\ \sin 0.1t \end{pmatrix} N.$$

Table 3
Parameters' values of OMFFC, PID and TDEC controllers.

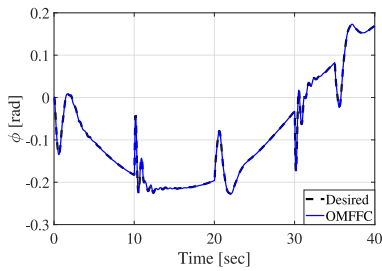
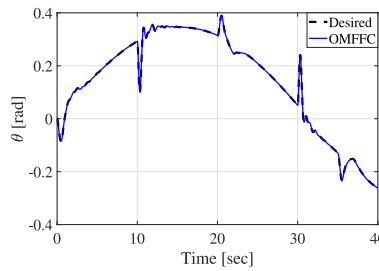
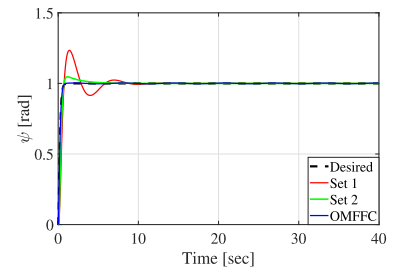
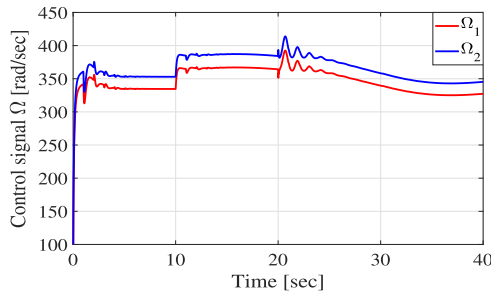
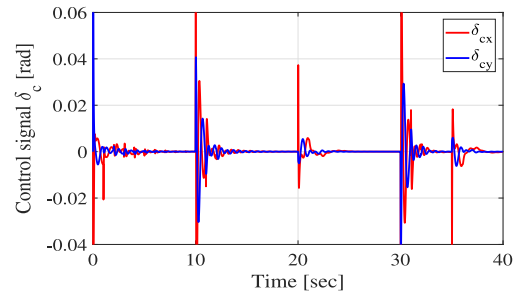
Controllers	Parameters	
	Position subsystems	Attitude subsystems
OMFFC	$\Lambda_{pi \in \{x,y\}} = 8.362$, $\Lambda_{pz} = 12.487$, $\Lambda_{li \in \{x,y\}} = 7.651$, $\Lambda_{lz} = 9.349$, $\Lambda_{Di \in \{x,y\}} = 7.364$, $\Lambda_{Dz} = 10.258$.	$\Lambda_{pi \in \{\phi,\theta\}} = 5.264$, $\Lambda_{p\psi} = 3.694$, $\Lambda_{Di \in \{\phi,\theta\}} = 3.264$, $\Lambda_{D\psi} = 3.115$, $\Lambda_{li \in \{\phi,\theta\}} = 4.759$, $\Lambda_{l\psi} = 3.276$.
PID	$k_{pi \in \{x,y,z\}} = 10$, $k_{li \in \{x,y,z\}} = 5$, $k_{Di \in \{x,y,z\}} = 7$.	$k_{pi \in \{\phi,\theta\}} = 6$, $k_{p\psi} = 4$, $k_{li \in \{\phi,\theta,\psi\}} = 4$, $k_{Di \in \{\phi,\theta,\psi\}} = 4$.
TDEC	$\Gamma_{pi \in \{x,y,z\}} = 10$, $\Gamma_{Di \in \{x,y,z\}} = 7$.	$\Gamma_{pi \in \{\phi,\theta\}} = 6$, $\Gamma_{p\psi} = 4$, $\Gamma_{Di \in \{\phi,\theta,\psi\}} = 4$.



(a) Position response in 3D space.



(b) Position response in X-Y plot.

Fig. 4. Coaxial rotor position response (test 1).(a) Roll (ϕ).(b) Pitch (θ).(c) Yaw (ψ).**Fig. 5.** Coaxial rotor attitude response (test1).(a) Control signals Ω_1 and Ω_2 .(b) Control signals δ_{cx} and δ_{cy} .**Fig. 6.** Control signals using OMFFC (test1).

The results of this section are shown in Figs. 7–9. Fig. 7 presents the position response in 3D and the X–Y plot comparison results obtained by applying the OMFFC, TDEC, and PID controllers to the coaxial rotor under external disturbances. This figure highlights the superiority of the proposed controller over the TDEC and the PID controllers due to the TDE technique and adaptive fuzzy logic compensator included. The roll, pitch, and yaw angle responses are plotted in Fig. 8. The yaw angle (ψ) response shows the effectiveness of the proposed control method

compared to the other controllers. The control inputs of the coaxial rotor are depicted in Fig. 9 whose evolutions show that the designed control approach succeeded in achieving the desired path despite the presence of external disturbances.

A quantitative comparison in the two previous operating modes (Tests 1 and 2) is also made between our proposed controller and the other control methods for both position and attitude subsystems based on RMSE and maximum absolute error (MaxAE) criteria. The results are listed in Table 4.

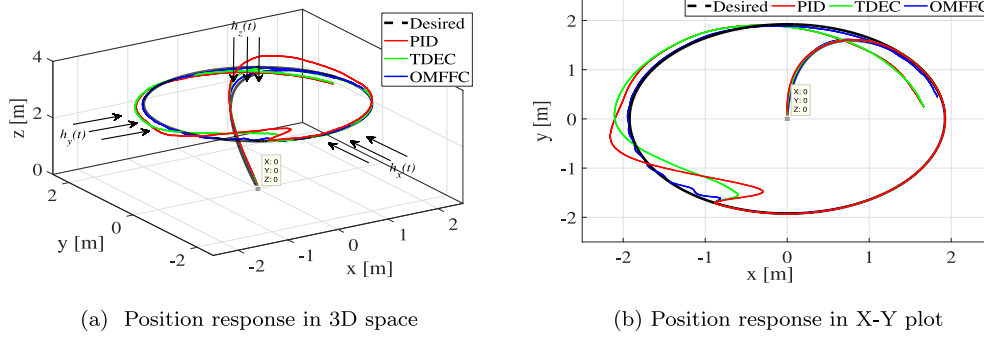


Fig. 7. Coaxial rotor position response (test 2).

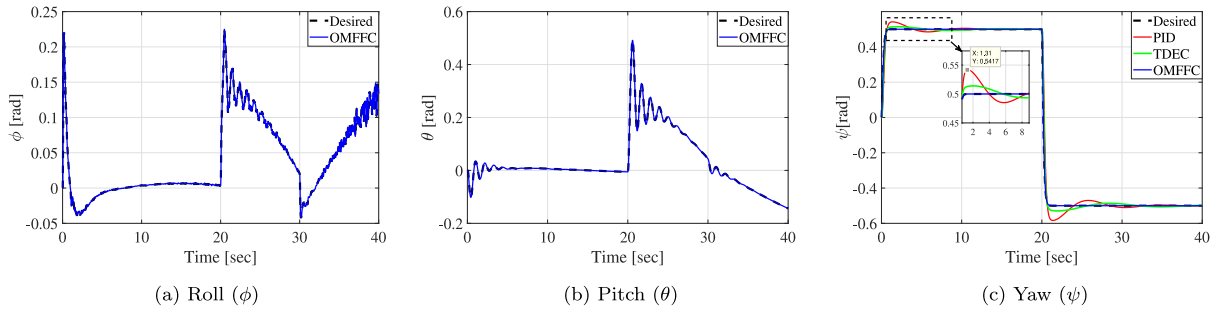


Fig. 8. Coaxial rotor attitude response (test 2).

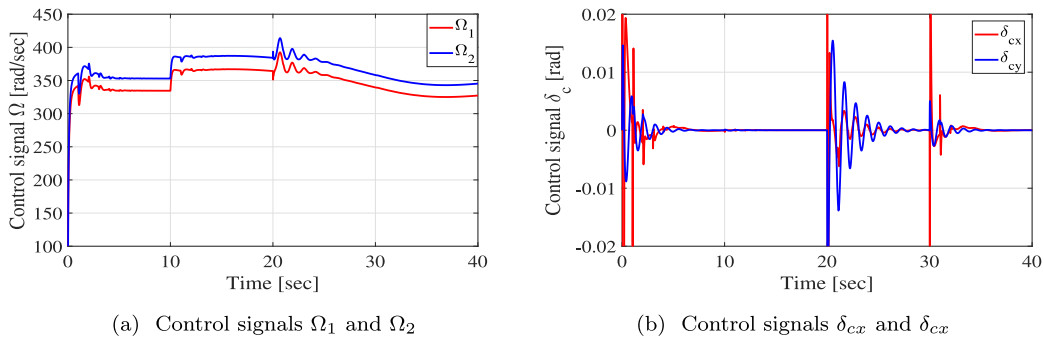


Fig. 9. Control signals using OMFFC (test 2).

Table 4
RMSE and MaxAE performance indexes.

Criterion	Test 1			Test 2		
	OMFFC	Set 1	Set 2	OMFFC	TEDC	PID
RMSE	7.51×10^{-4}	2.10×10^{-3}	2.80×10^{-3}	5.41×10^{-4}	1.80×10^{-3}	2.60×10^{-3}
MaxAE	1.50×10^{-1}	3.75×10^{-1}	2.68×10^{-1}	0.81×10^{-1}	2.19×10^{-1}	3.51×10^{-1}

From the obtained results, it is clear that the proposed OMFFC offers the smallest error values with respect to MFFC (Sets 1 and 2), TDEC and PID control methods which present the largest ones for almost all subsystems. It can be concluded that the system performances are better, when using the proposed OMFFC compared to the other controllers.

4.3. Test 3: Robustness to sensor noise and parametric variations

The robustness of the proposed controller was tested by considering sensing noise and by varying the system parameters from their nominal value. The performance of the coaxial rotor was then investigated in the case of realistic uncertain signals by

considering white Gaussian noise signals on velocities and accelerations of the coaxial rotor. Furthermore, in order to accommodate system degradation caused by the parameter uncertainties, mechanical characteristics of the coaxial rotor such as m , I_x and I_y are varied from the identified ones. The parameters variations and noise signals added to the coaxial system are listed below:

- uncertainties Δm (30%), ΔI_x and ΔI_y (35%) applied at 10, 20 and 30 s, respectively;
- Gaussian noises with 0.002 variance on position and attitude velocities, and 0.003 variance on the translation acceleration;

The desired trajectory is chosen as $x_d = 2(1 - \cos(0.2t))m$, $y_d = 2\sin(0.4t)m$, $z_d = 3$ m and $\psi_d = 1$ deg, and the initial

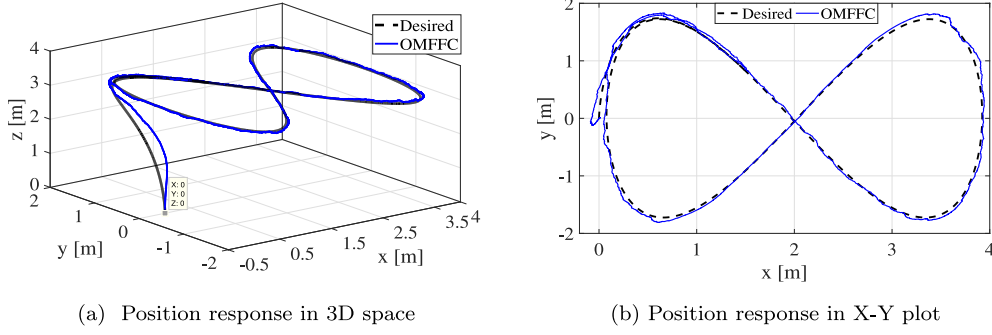


Fig. 10. Coaxial rotor position response (test 3).

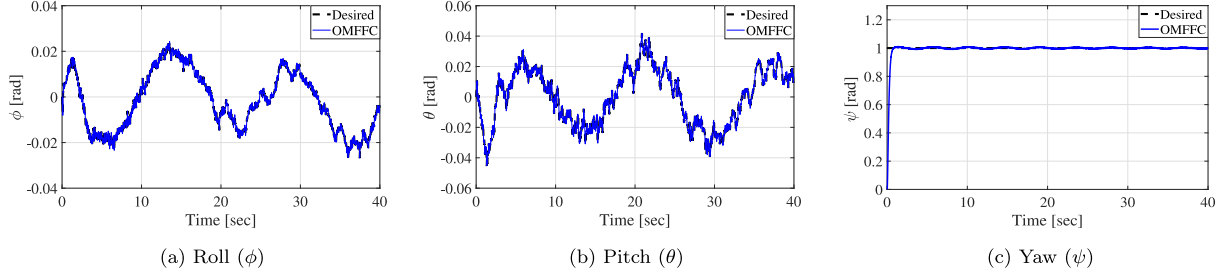


Fig. 11. Coaxial rotor attitude response (test 3).

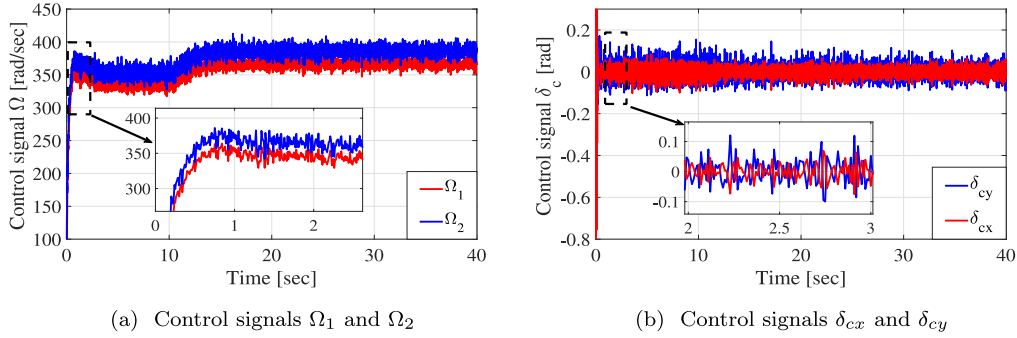


Fig. 12. Control signals using OMFFC (test 3).

conditions are chosen the same as of previous tests 1 and 2. This test describes a complex real-world scenario that may be important for future experimental validation. Figs. 10–12 depict simulation results of the system when sensors of the coaxial rotor are noisy throughout his flight mission. Fig. 10, presenting its position which shows that the assigned navigational task is successful, and the reference trajectory is tracked with high accuracy achieved even with uncertainties and noise. The attitude tracking performances are practically the same as obtained in tests 1 and 2, see Fig. 11.

The response curves of the control inputs Ω_1 , Ω_2 , δ_{cx} and δ_{cy} are shown in Fig. 12. The amplitude of the control signals becomes larger than noise free but remains in their eligible regions. The control signals are adjusted by adaptive fuzzy control terms so that can compensate the effects caused by noisy measurements and the loss of physical characteristics. The simulation results of all flight tests illustrate the effectiveness and the strong robustness of the OMFFC approach against disturbances, parameter uncertainties and sensors noise.

5. Conclusion

In this paper, a novel time-delay estimation (TDE) based approach has been proposed to design optimal model-free control

for position and attitude control of a coaxial rotor drone. This method relies on the use of the TDE technique to estimate unknown dynamic functions of the coaxial rotor and an additional robust adaptive fuzzy logic control term for the effective estimation of error compensation. These facts not only allow the shortcomings of the model-based approaches to be overcome, thus improving the robustness of the controller, but also the computational complexity of the design conditions to be reduced. Moreover, the control design is formalized using the Lyapunov theory where the optimal controller gains are then computed by resolving an optimization problem through the exploitation of a metaheuristic flower pollination-based optimization algorithm aiming to minimize the tracking errors while guaranteeing the closed-loop stability properties of the whole system. Furthermore, the control structure is relatively simple with low-complexity design conditions. Extensive numerical results have been presented to illustrate the best performance of the proposed control algorithm in comparison with standard PID and time-delay estimation-based controllers under external disturbances. To go even further, the control approach was also validated taking the noise and parametric uncertainties into consideration. Thereafter, error quantification was proposed to evaluate the quality of the tracking. Future research should consider a model-free fault-tolerant control strategy using the optimal adaptive fuzzy system

to estimate and offset faults in the system. It should also focus on the real-time validation of the proposed method with our experimental platform under development.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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